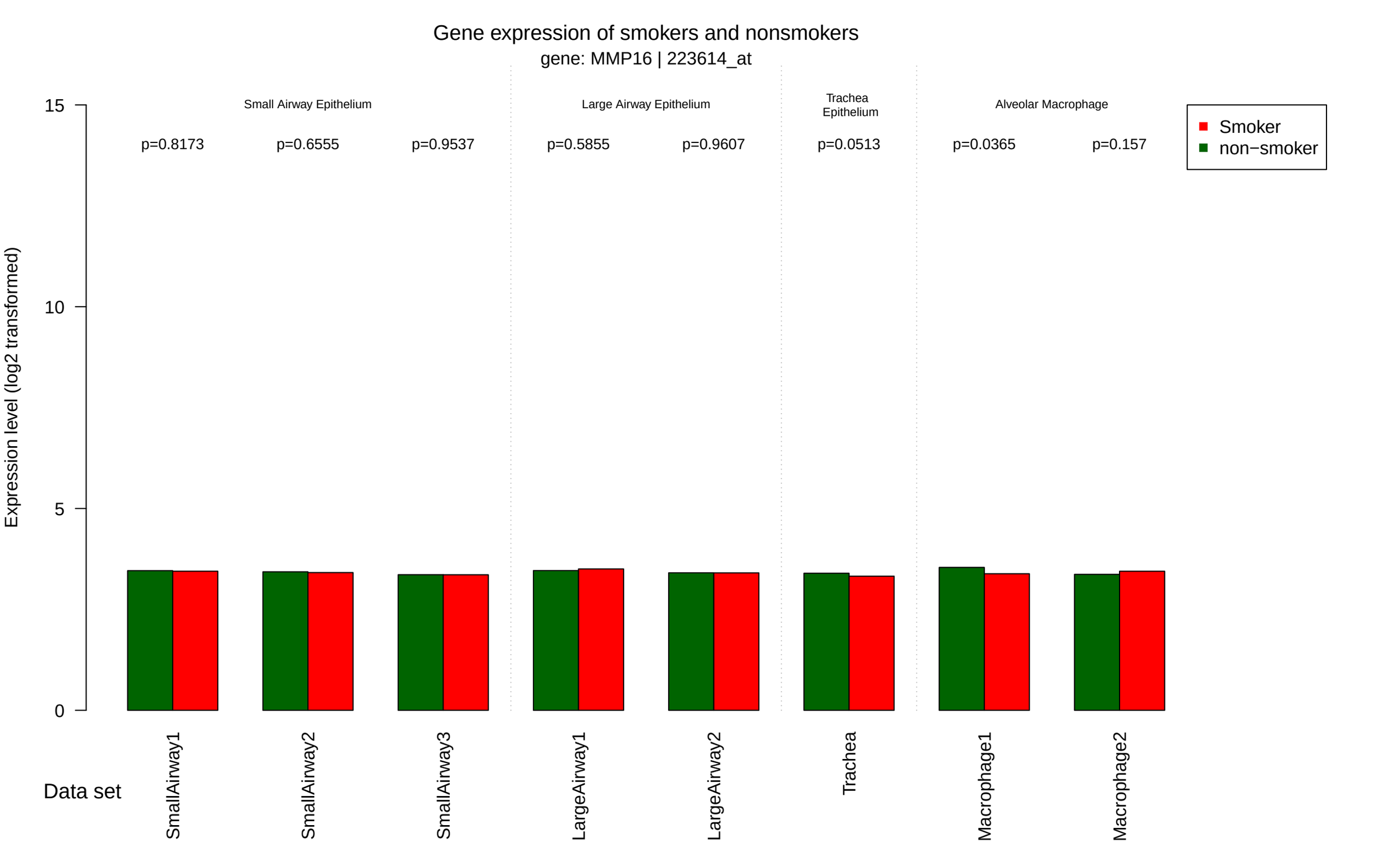
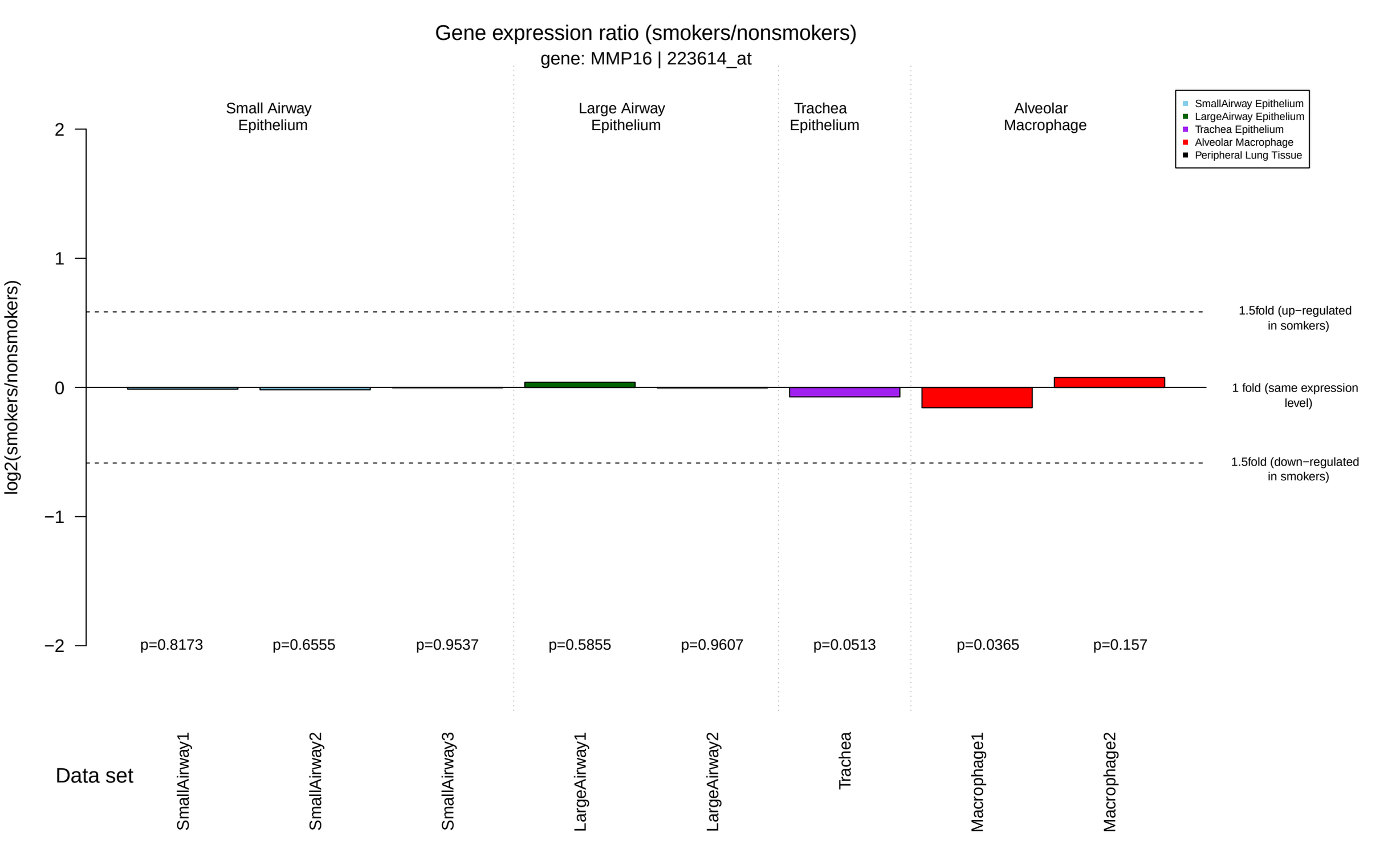
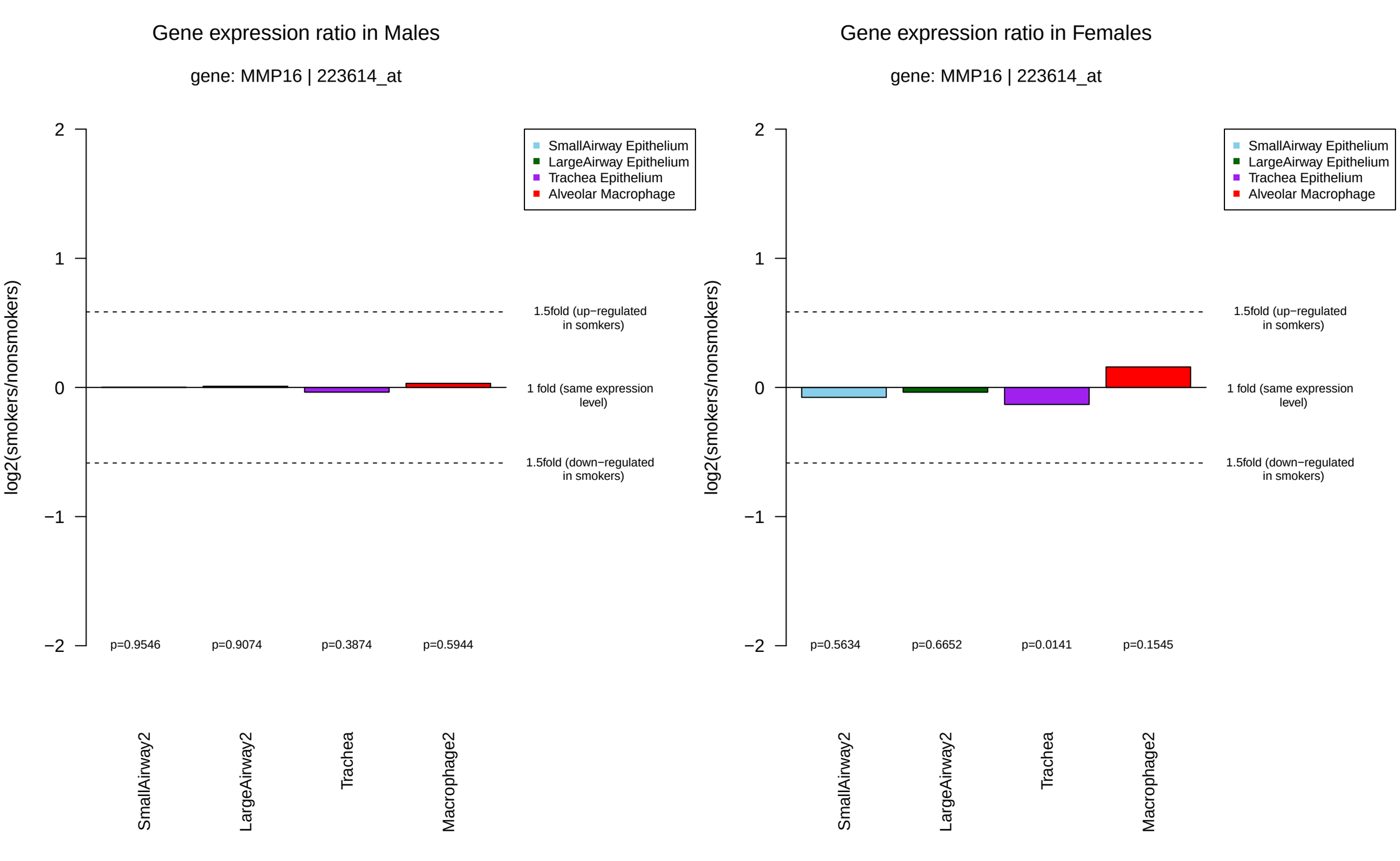
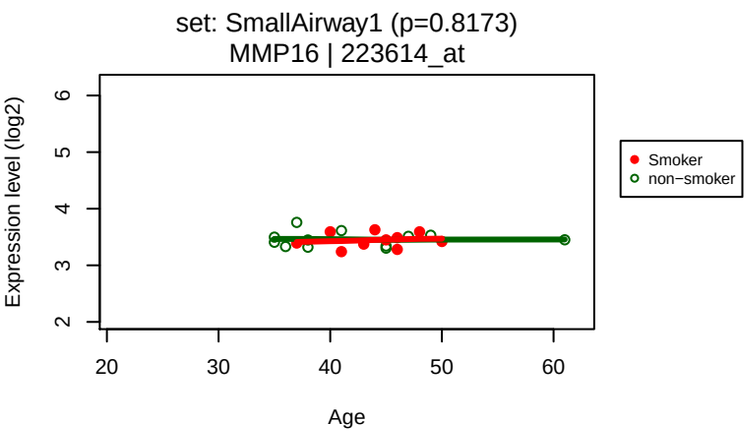
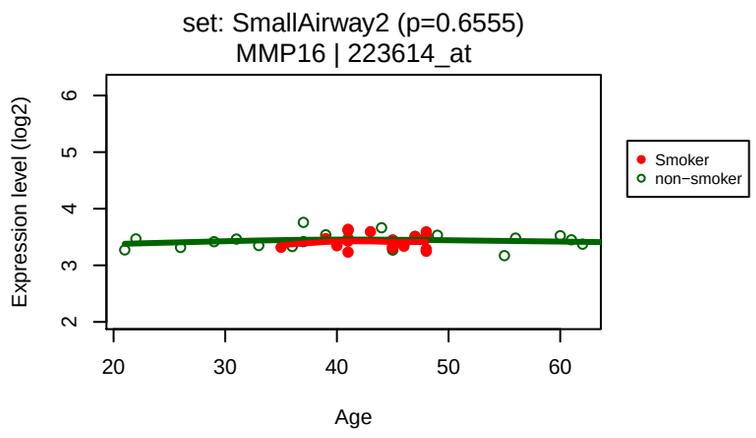
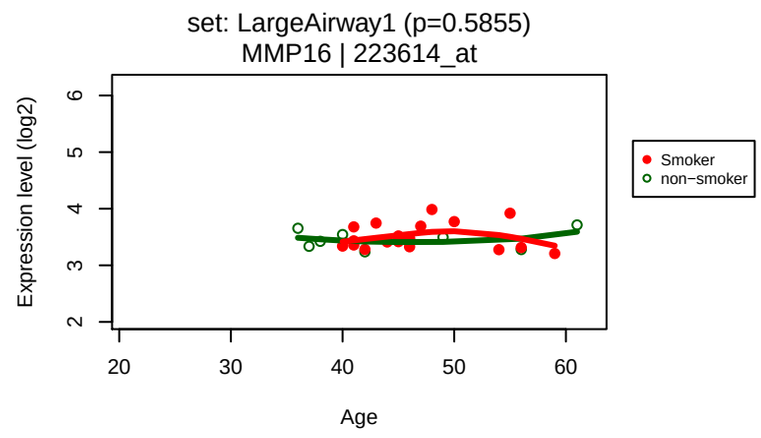

Gene expression of smokers and nonsmokers
gene: MMP16 | 223614_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8173
p=0.6555
p=0.9537
p=0.5855
p=0.9607
p=0.0513
p=0.0365
p=0.157
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

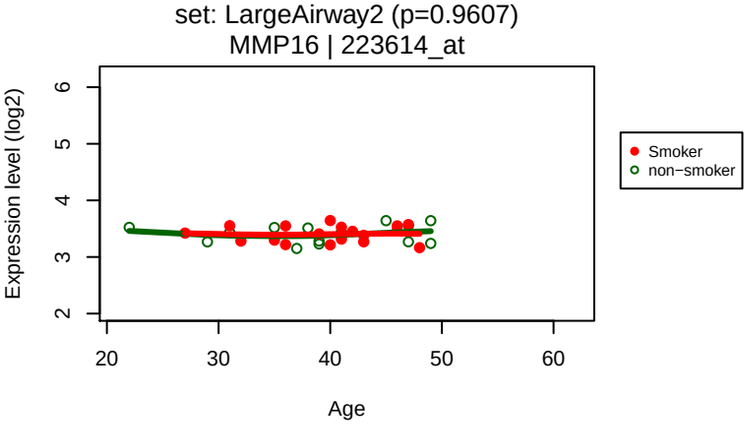

Gene expression ratio (smokers/nonsmokers)
gene: MMP16 | 223614_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8173
p=0.6555
p=0.9537
p=0.5855
p=0.9607
p=0.0513
p=0.0365
p=0.157
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

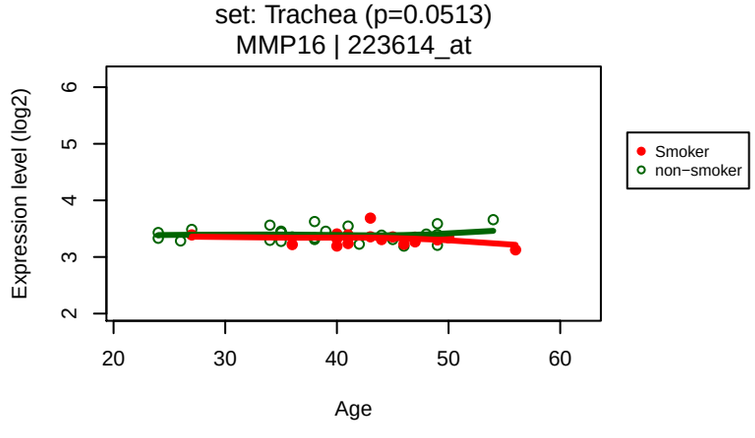

Gene expression ratio in Males
gene: MMP16 | 223614_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9546
p=0.9074
p=0.3874
p=0.5944
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MMP16 | 223614_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5634
p=0.6652
p=0.0141
p=0.1545
SmallAirway2
LargeAirway2
Trachea
Macrophage2

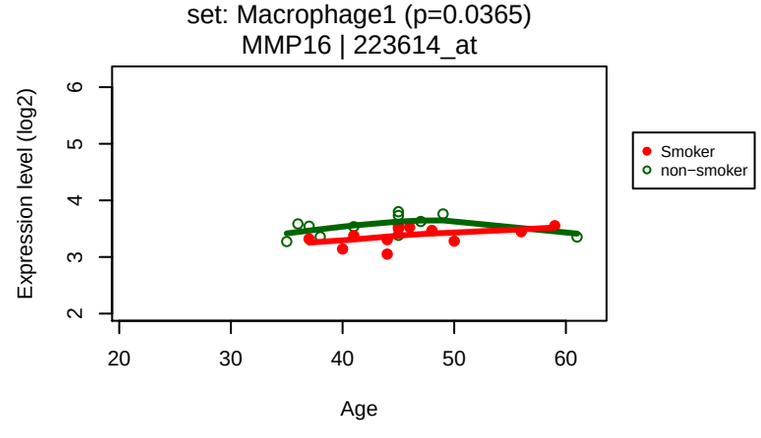

set: SmallAirway1 (p=0.8173)
MMP16 | 223614_at
Expression level (log2)
Age
Smoker
non−smoker

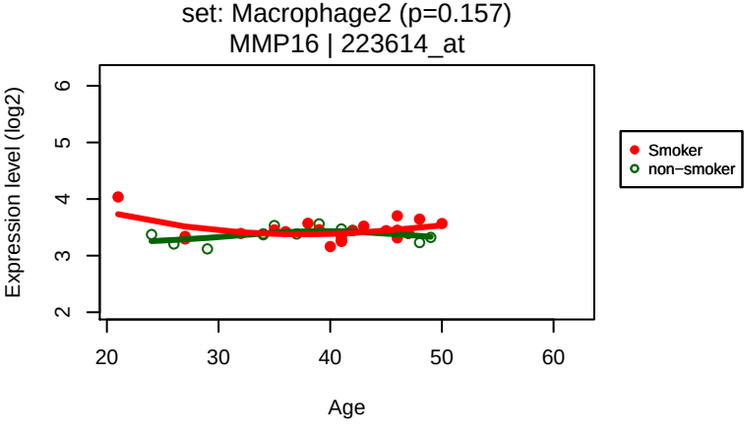

set: SmallAirway2 (p=0.6555)
MMP16 | 223614_at
Expression level (log2)
Age
Smoker
non−smoker

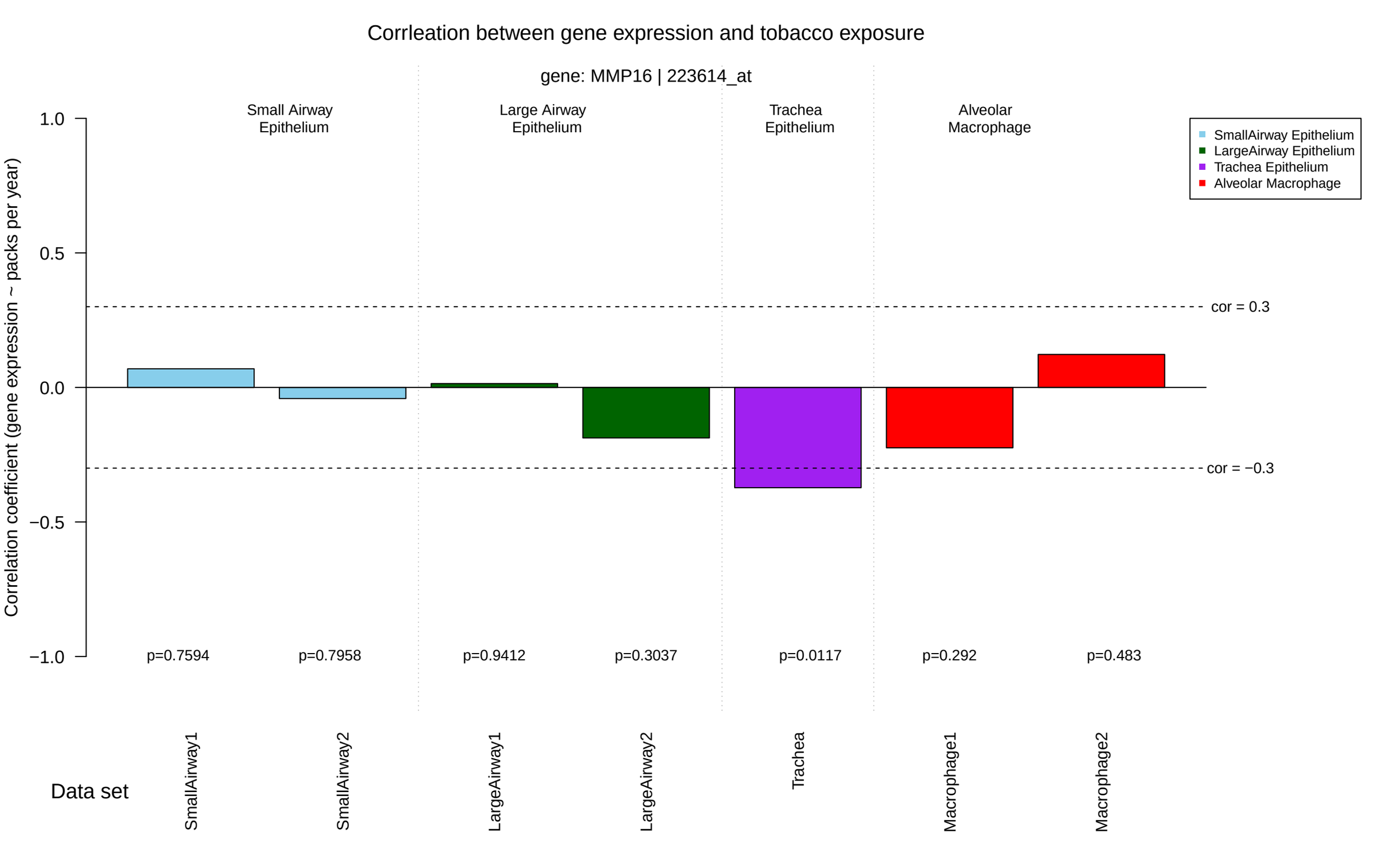

set: LargeAirway1 (p=0.5855)
MMP16 | 223614_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9607)
MMP16 | 223614_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0513)
MMP16 | 223614_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0365)
MMP16 | 223614_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.157)
MMP16 | 223614_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MMP16 | 223614_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7594
p=0.7958
p=0.9412
p=0.3037
p=0.0117
p=0.292
p=0.483
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2